SAT Math Mastery · Extra Practice

Thin Skills: Statistics, Probability, Functions and Similarity

Time about 2 hours: Part A in class, Part B independently.

Topics evaluating claims · margin of error · probability · two-variable data · composition and transformations of functions · sums and products of roots · similar triangles and area ratios

Name / Date _____ _____

PART A · IN CLASS — 25 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos (or a calculator), one expression per line; tap a grey dot to read an intersection or a vertex.

1. HARD · NO CALCULATOR NEEDED

A researcher recruited 120 volunteers who responded to an online advertisement. She randomly assigned 60 of them to study with method X and 60 to study with method Y for a month. The method X group had a higher mean score on a final test, and the difference was larger than would be expected from random assignment alone. Which conclusion is best supported by the study?

(A) The study provides no evidence about either method.

(B) Method X caused higher scores for volunteers like those in the study. ✓

(C) Method X is associated with higher scores, but no causal conclusion is possible.

(D) Method X causes higher scores for all students.

1. Count the people who were randomly assigned: Type `60+60`: it returns 120 volunteers, so every participant was assigned at random.
2. Check the group sizes: Type `60/120`: it returns 0.5 went to each method. Random assignment supports a cause-and-effect conclusion.
3. But the 120 were volunteers who answered an advertisement, not a random sample of all students, so the conclusion applies only to volunteers like them.
4. Best supported: method X caused higher scores for volunteers like those in the study.

Answer: (B)

2. HARD · NO CALCULATOR NEEDED

A random sample of 500 adults was selected from a large city, and each adult reported how many hours per week they exercise and how many hours they sleep each night. The adults who exercise more tend to sleep more. Which conclusion is best supported by this study?

(A) Exercising more causes adults in the city to sleep more.

(B) Sleeping more causes adults in the city to exercise more.

(C) Among the 500 adults only, exercise and sleep are associated; nothing can be said about the city.

(D) Among adults in the city, exercise and sleep are positively associated, but exercise has not been shown to cause more sleep. ✓

1. Random selection: the 500 adults were chosen at random from the city, so the result generalizes to adults in the city (no calculation is needed here).
2. Nobody was assigned to exercise more or less, so the study is observational and can show association only.
3. Reading the data: those who exercise more tend to sleep more, a positive association. Choices A and B claim a cause; C refuses to generalize.
4. Best supported: a positive association among adults in the city, with no cause shown.

Answer: (D)

3. HARD · NO CALCULATOR NEEDED

In a poll of a random sample of voters, 51% said they support a proposed tax. The margin of error was 2.5% at the 95% confidence level. Based on the poll, which statement is best supported?

(A) A majority of all voters support the tax.

(B) It is plausible that a majority support the tax, but less than half supporting it is also plausible. ✓

(C) Fewer than half of all voters support the tax.

(D) Exactly 51% of all voters support the tax.

1. Lower end of the plausible values: Type `51-2.5` : it returns 48.5.
2. Upper end: Type `51+2.5` : it returns 53.5.
3. The interval 48.5 to 53.5 contains 50, so both more than half and less than half are plausible.
4. Best supported: it is plausible that a majority support the tax, but less than half is also plausible.

Answer: (B)

4. HARD · NO CALCULATOR NEEDED

A random sample of 500 residents of Town P gave an estimate of 30% for the percent who favor a new park, with a margin of error of 4%. A random sample of 500 residents of Town Q gave an estimate of 38%, also with a margin of error of 4%. Which statement is best supported?

(A) It is plausible that the percent in favor is the same in both towns. ✓

(B) The percent in favor in Town P is exactly 4 points less than that in Town Q.

(C) Town Q definitely has a greater percent in favor than Town P.

(D) It is not possible for the percents to be equal.

1. Town P: Type `30-4`: it returns 26 to Type `30+4`: it returns 34 percent.
2. Town Q: Type `38-4`: it returns 34 to Type `38+4`: it returns 42 percent.
3. The intervals share the value 34, so the same percent in both towns is plausible.
4. Best supported: it is plausible that the percent in favor is the same in both towns.

Answer: (A)

5. HARD · NO CALCULATOR NEEDED

A school has 1,200 seniors and 2,800 students in other grades. A random sample of 300 seniors was surveyed, and an estimated 62% plan to attend college, with a margin of error of 5%. Which statement is best supported by the survey?

(A) Exactly 62% of the seniors at the school plan to attend college.

(B) Between 57% and 67% of the 300 surveyed seniors plan to attend college.

(C) Between 57% and 67% of all students at the school plan to attend college.

(D) It is plausible that between 57% and 67% of the seniors at the school plan to attend college. ✓

1. Interval for the surveyed group: Type `62-5`: it returns 57 to Type `62+5`: it returns 67 percent.
2. Seniors are what share of the school? Type `1200/(1200+2800)`: it returns 0.3, and the sample came only from them.
3. So the interval describes seniors at the school, not all students, and it is a range of plausible values, not an exact value.
4. Best supported: it is plausible that between 57% and 67% of the seniors at the school plan to attend college.

Answer: (D)

6. HARD · DESMOS

Events A and B are independent. The probability of A is 0.4, and the probability that at least one of A and B occurs is 0.7. What is the probability of B?

1. For independent events $P(A \text{ or } B) = P(A) + P(B) - P(A)P(B)$. Type `y=0.4+x-0.4x` and `y=0.7` ($x$ stands for $P(B)$).
2. Tap the intersection: (0.5, 0.7). Its x-value is $P(B)$.
3. Check: Type `0.4+0.5-0.4*0.5`: it returns 0.7.
4. $P(B) = 0.5$.

Answer: 0.5 **(or** $\frac{1}{2}$**)**

7. Hard · Desmos

Two fair six-sided dice are rolled. Given that at least one of the dice shows an even number, what is the probability that the sum of the two dice is 8?

1. Outcomes with at least one even die: Type `36-3*3` : it returns 27 (remove the 3×3 all-odd pairs).
2. Sum 8 with an even die: $(2,6), (6,2), (4,4)$, so 3 outcomes. ($(3,5)$ and $(5,3)$ are all-odd.)
3. Probability: Type `3/27` : it returns 0.1111.
4. Answer: $\frac{1}{9}$ (enter 1/9).

Answer: $\frac{1}{9}$ **(or** 0.1111**)**

8. Hard · Desmos

In a town, 30% of the residents are over 60 years old. Of the residents over 60, 80% vote in elections; of the residents who are not over 60, 50% vote. A voter is chosen at random from all residents who vote. What is the probability that the voter is over 60?

1. Voters over 60: Type `0.3*0.8` : it returns 0.24.
2. Voters not over 60: Type `0.7*0.5` : it returns 0.35.
3. Share of voters who are over 60: Type `0.24/(0.24+0.35)` : it returns 0.4068.
4. Answer: $\frac{24}{59}$ (enter 24/59).

Answer: $\frac{24}{59}$ **(or** 0.4067 **or** 0.4068**)**

9. Hard · Desmos

The table shows values of x and y for a data set.

x	0	1	2	3	4
y	8	12	18	27	40.5

Which equation fits all five pairs?

(A) $y = 8(1.5)^x$ ✓

(B) $y = 4x + 8$

(C) $y = 2x^2 + 2x + 8$

(D) $y = 8(1.2)^x$

1. Open a table with $x_1 = 0, 1, 2, 3, 4$ and $y_1 = 8, 12, 18, 27, 40.5$. Type `y_1~a*b^{x_1}` : Desmos returns $a = 8$ and $b = 1.5$ (an exponential fit).
2. Test (A) at $x = 4$: Type `8*1.5^4` : it returns 40.5, matching 40.5.
3. Test (C) at $x = 2$: Type `2*2^2+2*2+8` : it returns 20, not 18; test (D) at $x = 1$: Type `8*1.2^1` : it returns 9.6, not 12; test (B) at $x = 2$: Type `4*2+8` : it returns 16, not 18.
4. Only $y = 8(1.5)^x$ fits all five pairs: (A).

Answer: (A)

10. HARD · DESMOS

Ten data points lie exactly on the line $y = 2x + 1$, at $x = 1, 2, \ldots, 10$. A new data point $(11, 40)$, far above that line, is added. How does the least-squares line for all eleven points compare with the line $y = 2x + 1$?

(A) Smaller slope and greater y-intercept

(B) The same slope and the same y-intercept

(C) Greater slope and smaller y-intercept ✓

(D) Greater slope and greater y-intercept

1. The old line predicts Type `2*11+1`: it returns 23 at $x = 11$, so the new point is Type `40-23`: it returns 17 above the line.
2. Open a table with $x_1 = 1, 2, \ldots, 11$ and $y_1 = 3, 5, \ldots, 21, 40$. Type `y_1~mx_1+b`: Desmos returns $m = 2.773$ and $b = -2.091$.
3. Compare with $y = 2x + 1$: the slope 2.773 is greater than 2 and the intercept -2.091 is smaller than 1.
4. Answer: (C) greater slope and smaller y-intercept.

Answer: (C)

11. HARD · DESMOS

The residuals for the first four of five data points about a least-squares line are -2, 3, 1 and -4. What is the residual for the fifth point?

1. Residuals add to zero for a least-squares line. Add the four known ones: Type `-2+3+1-4`: it returns -2.
2. The fifth residual cancels that: Type `0-(-2)`: it returns 2.
3. Check: Type `-2+3+1-4+2`: it returns 0.
4. Answer: 2.

Answer: 2

12. HARD · DESMOS

The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = -2$ and $x = 5$. The function g is defined by $g(x) = f(2x)$. What are the x-intercepts of the graph of $y = g(x)$?

(A) $x = -2$ and $x = \dfrac{5}{2}$

(B) $x = -4$ and $x = 10$

(C) $x = -1$ and $x = \dfrac{5}{2}$ ✓

(D) $x = 0$ and $x = 3$

1. Pick a sample f with zeros -2 and 5: $f(x) = (x+2)(x-5)$. Then $g(x) = f(2x) = (2x+2)(2x-5)$. Type `y=(2x+2)(2x-5)`.
2. Tap the two x-intercepts: $(-1, 0)$ and $(2.5, 0)$.
3. Check: Type `-2/2`: it returns -1 and Type `5/2`: it returns 2.5, the old zeros divided by 2.
4. Answer: (C) $x = -1$ and $x = \dfrac{5}{2}$.

Answer: (C)

13. HARD · DESMOS

The functions f and g are defined for the values in the table.

x	1	2	3	4
$f(x)$	3	4	1	2
$g(x)$	4	1	2	3

What is the value of $g\left(f^{-1}(g(3))\right)$?

1. Open a table with $x_1 = 1, 2, 3, 4$, $f_1 = 3, 4, 1, 2$, $g_1 = 4, 1, 2, 3$. First $g(3)$: type `(g_1[3])` : it returns 2.
2. Now find x with $f(x) = 2$: look down the f column, $f(4) = 2$, so type `(f_1[4])` : it returns 2, and $f^{-1}(2) = 4$.
3. Finally $g(4)$: type `(g_1[4])` : it returns 3.
4. Answer: 3.

Answer: 3

14. HARD · DESMOS

The function f is odd, which means $f(-x) = -f(x)$ for every x. Which of the following functions must be even, meaning $h(-x) = h(x)$ for every x?

(A) $[f(x)]^2$ ✓

(B) $f(f(x))$

(C) $f(x) + 1$

(D) $f(x) + x$

1. Pick an odd function to test, $f(x) = x^3$, and compare $x = 2$ with $x = -2$. (A): Type `(2^3)^2` : it returns 64 and Type `((-2)^3)^2` : it returns 64: equal, so even.
2. (B) $f(f(x))$: Type `(2^3)^3` : it returns 512 and Type `((-2)^3)^3` : it returns -512: opposites, so odd.
3. (C) $f(x) + 1$: Type `2^3+1` : it returns 9 and Type `(-2)^3+1` : it returns -7: not equal. (D) $f(x) + x$: Type `2^3+2` : it returns 10 and Type `(-2)^3-2` : it returns -10: opposites.
4. Only (A) $[f(x)]^2$ must be even.

Answer: (A)

15. HARD · DESMOS

Let $f(x) = \sqrt{x - 1}$ and $g(x) = x^2 - 5$. Which describes all real values of x for which $f(g(x))$ is defined?

(A) $x \geq 1$

(B) $-\sqrt{6} \leq x \leq \sqrt{6}$

(C) all real numbers

(D) $x \leq -\sqrt{6}$ **or** $x \geq \sqrt{6}$ ✓

1. $f(g(x)) = \sqrt{(x^2 - 5) - 1} = \sqrt{x^2 - 6}$. Type `y=sqrt(x^2-6)` .
2. Desmos draws the curve only where it is defined. Tap the x-intercepts: $(-2.4495, 0)$ and $(2.4495, 0)$.
3. Check: Type `sqrt(6)` : it returns 2.4495, and the curve exists for $x \leq -\sqrt{6}$ and $x \geq \sqrt{6}$ only (nothing between).
4. Answer: (D) $x \leq -\sqrt{6}$ or $x \geq \sqrt{6}$.

Answer: (D)

16. HARD · DESMOS

The graph of $y = f(x)$ passes through the point $(4, -2)$. Which point must lie on the graph of $y = 3f(x-1) + 5$?

(A) $(5, 11)$ **(B) $(5, -1)$ ✓**

(C) $(3, -1)$ (D) $(3, 11)$

1. $f(x-1)$ moves the point right 1: Type `4+1` : it returns 5.
2. Then y becomes $3(-2) + 5$: Type `3(-2)+5` : it returns -1.
3. The point is $(5, -1)$. Check with the other options: Type `3(-2)+5` : it returns -1 is not 11, so (A) and (D) fail; $x = 3$ would need a left shift.
4. Answer: (B) $(5, -1)$.

Answer: (B)

17. HARD · DESMOS

The equation $2x^2 - 7x + k = 0$, where k is a constant, has two real solutions r and s with $r^2 + s^2 = 9$. What is the value of k?

1. By Vieta $r + s = \frac{7}{2}$ and $rs = \frac{k}{2}$, and $r^2 + s^2 = (r+s)^2 - 2rs$, so $k = (r+s)^2 - 9$: Type `(7/2)^2-9` : it returns 3.25.
2. Check the roots for $k = 3.25$: Type `(7+sqrt(49-8*3.25))/4` : it returns 2.949 and Type `(7-sqrt(49-8*3.25))/4` : it returns 0.551.
3. Their squares add to: Type `((7+sqrt(49-8*3.25))/4)^2+((7-sqrt(49-8*3.25))/4)^2` : it returns 9.
4. Answer: $k = \frac{13}{4}$ (enter 13/4).

Answer: $\frac{13}{4}$ **(or** 3.25**)**

18. HARD · DESMOS

What is the sum of all real solutions of $2^{2x} - 6 \cdot 2^x + 8 = 0$?

1. Type `y=2^(2x)-6*2^x+8` .
2. Tap the x-intercepts: (1, 0) and (2, 0).
3. Add the solutions: Type `1+2` : it returns 3.
4. Answer: 3.

Answer: 3

19. HARD · DESMOS

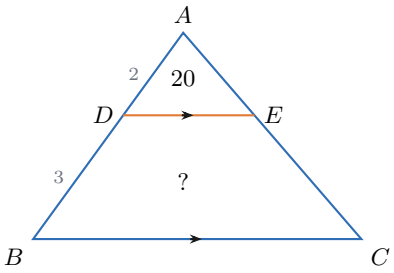
Triangle ABC is similar to triangle DEF. The area of triangle ABC is 36 and the area of triangle DEF is 100. The perimeter of triangle ABC is 24. What is the perimeter of triangle DEF?

1. Scale factor = square root of the area ratio: Type `sqrt(100/36)` : it returns 1.6667.
2. Perimeters scale by the scale factor itself: Type `24*sqrt(100/36)` : it returns 40.
3. Check with the squares: Type `(24*sqrt(100/36)/24)^2` : it returns 2.7778 equals Type `100/36` : it returns 2.7778.
4. Answer: 40.

Answer: 40

20. HARD · DESMOS

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD : DB = 2 : 3$ and the area of triangle ADE is 20, what is the area of quadrilateral $DBCE$?



1. $AD : AB = 2 : 5$, so the scale factor is Type `2/5` : it returns 0.4.

2. Area of ABC: Type `20/(2/5)^2` : it returns 125.

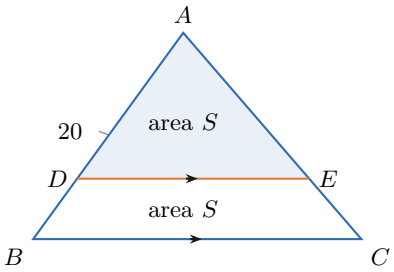
3. Quadrilateral $DBCE$: Type `125-20` : it returns 105.

4. Answer: 105.

Answer: 105

21. HARD · DESMOS

In triangle ABC, point D is on $\overline{AB}$ and point E is on $\overline{AC}$ with $\overline{DE} \parallel \overline{BC}$. The segment $\overline{DE}$ divides the triangle into two regions of equal area. If $AB = 20$, what is the length of $\overline{AD}$, to the nearest tenth?



1. $\triangle ADE$ has half the area of $\triangle ABC$, so $\dfrac{AD}{AB} = \sqrt{\frac{1}{2}}$: Type `sqrt(1/2)` : it returns 0.7071.

2. Multiply by $AB = 20$: Type `20*sqrt(1/2)` : it returns 14.1421.

3. Check: Type `(20*sqrt(1/2)/20)^2` : it returns 0.5 is the area ratio $\dfrac{1}{2}$.

4. Answer: 14.1.

Answer: 14.1

22. HARD · DESMOS

A square pyramid has a base with area 144. A plane parallel to the base cuts the pyramid at a point one-third of the way from the apex to the base, forming a smaller square cross-section. What is the area of the cross-section?

1. The scale factor from apex to the cut is $\frac{1}{3}$. Area factor: Type `(1/3)^2` : it returns 0.1111.
2. Cross-section area: Type `144*(1/3)^2` : it returns 16.
3. Check with side lengths: base side Type `sqrt(144)` : it returns 12, cut side Type `sqrt(144)/3` : it returns 4, area Type `(sqrt(144)/3)^2` : it returns 16.
4. Answer: 16.

Answer: 16

23. HARD · DESMOS

The graph of $y = f(x)$ has x-intercepts at $x = 1$ and $x = 6$. The function g is defined by $g(x) = f(3 - x)$. What is the greatest x-intercept of the graph of $y = g(x)$?

1. Take a sample f with zeros 1 and 6, $f(x) = (x - 1)(x - 6)$, so $g(x) = f(3 - x) = (3 - x - 1)(3 - x - 6)$. Type `y=(3-x-1)(3-x-6)` .
2. Tap the x-intercepts: $(-3, 0)$ and $(2, 0)$.
3. Check: Type `3-1` : it returns 2 and Type `3-6` : it returns -3.
4. Greatest x-intercept: 2.

Answer: 2

24. HARD · DESMOS

The solutions of $x^2 - 5x + 2 = 0$ are r and s. What is the value of $\frac{r}{s} + \frac{s}{r}$?

1. Type `y=x^2-5x+2` and tap the x-intercepts: (0.4384, 0) and (4.5616, 0). These are s and r.
2. Type Type `((5+sqrt(17))/2)/((5-sqrt(17))/2)+((5-sqrt(17))/2)/((5+sqrt(17))/2)` : it returns 10.5.
3. Check with Vieta: Type `(5^2-2*2)/2` : it returns 10.5.
4. Answer: $\frac{21}{2}$ (enter 21/2).

Answer: $\frac{21}{2}$ **(or** 10.5**)**

25. HARD · DESMOS

The polynomial $p(x) = x^3 - 7x^2 + 14x - 8$ has three real zeros. What is the sum of the squares of the zeros?

1. Type `y=x^3-7x^2+14x-8` .
2. Tap the three x-intercepts: (1, 0), (2, 0), (4, 0).
3. Sum of the squares: Type `1^2+2^2+4^2` : it returns 21.
4. Answer: 21.

Answer: 21

PART B · SELF-ASSESSMENT — 20 PROBLEMS

26. HARD · DESMOS

A random sample of students at a university reported a mean of 3.2 hours of study per day, with a margin of error of 0.4 hours. The university has 18,000 students. Based on the sample, what is the greatest total number of study hours per day that is plausible for the whole student body?

1. Greatest plausible mean: Type `3.2+0.4` : it returns 3.6 hours per day.
2. Multiply by the number of students: Type `3.6*18000` : it returns 64,800.
3. Check the sample-mean total for comparison: Type `3.2*18000` : it returns 57,600 (this ignores the margin of error, the trap).
4. The greatest plausible total is 64,800 hours per day.

Answer: 64,800

27. HARD · DESMOS

To estimate the number of fish in a lake, a biologist caught and tagged 150 fish and released them. A week later she caught a random sample of 60 fish from the lake, and 9 of them were tagged. Assuming the sample is representative and the tagged fish mixed evenly, what is the best estimate of the number of fish in the lake?

1. Tagged fraction in the sample: Type `9/60` : it returns 0.15.
2. That fraction should equal $\frac{150}{N}$. Type `y=150/x` and `y=9/60` , tap the intersection: (1,000, 0.15).
3. Check: Type `150/(9/60)` : it returns 1,000 fish.
4. Estimated number of fish: 1000.

Answer: 1000

28. HARD · DESMOS

A company's customers were asked whether they renewed their contract. The results are shown in the table.

	Renewed	Did not renew
Plan A	84	36
Plan B	45	35

If one of the customers who did not renew is chosen at random, what is the probability that the customer was on Plan B?

1. Customers who did not renew: Type `36+35` : it returns 71.
2. Those on Plan B: 35, so type Type `35/71` : it returns 0.493.
3. Compare the trap (all 200 customers): Type `35/200` : it returns 0.175, which is wrong.
4. Probability $\frac{35}{71} \approx 0.493$ (enter 35/71).

Answer: $\frac{35}{71}$ **(or** 0.4929 **or** 0.4930**)**

29. Hard · Desmos

A bag holds 5 red marbles and 3 blue marbles. Two marbles are drawn at random, one after the other, without replacement. What is the probability that the two marbles are the same color?

(A) $\frac{15}{28}$ **(B) $\frac{13}{28}$ ✓**

(C) $\frac{5}{14}$ (D) $\frac{17}{32}$

1. Both red: Type `(5/8)(4/7)` : it returns 0.3571.

2. Both blue: Type `(3/8)(2/7)` : it returns 0.1071.

3. Add: Type `(5/8)(4/7)+(3/8)(2/7)` : it returns 0.4643.

4. Match with the options: Type `13/28` : it returns 0.4643, so the answer is $\frac{13}{28}$.

Answer: (B)

30. Hard · Desmos

A line of best fit for a data set is $\hat{y} = 3.2x + 14$. For the data point with $x = 10$, the actual value is $y = 51$. What is the residual for this point (actual value minus predicted value)?

1. Type `y=3.2x+14` and read the predicted value at $x = 10$: (10, 46).

2. Check: Type `3.2*10+14` : it returns 46.

3. Residual = actual − predicted: Type `51-46` : it returns 5.

4. Answer: 5.

Answer: 5

31. Hard · Desmos

A least-squares line for a data set is $\hat{y} = 1.5x + 4$. The mean of the x-values is 10. What is the mean of the y-values?

1. Type `y=1.5x+4` . The least-squares line passes through the point of means $(\bar{x}, \bar{y})$.

2. Read the line at $x = 10$: (10, 19).

3. Check: Type `1.5*10+4` : it returns 19.

4. Answer: $\bar{y} = 19$.

Answer: 19

32. HARD · DESMOS

The function f is defined by $f(x) = \dfrac{2x+1}{x-3}$ for $x \neq 3$. What is the value of $f^{-1}(5)$?

1. $f^{-1}(5)$ is the x with $f(x) = 5$. Type `y=(2x+1)/(x-3)` and `y=5`.

2. Tap the intersection on the right branch: (5.3333, 5). Its x-value is $f^{-1}(5)$.

3. Check: Type `16/3`: it returns 5.3333.

4. Answer: $\dfrac{16}{3}$ (enter 16/3).

Answer: $\dfrac{16}{3}$ **(or** 5.333**)**

33. HARD · DESMOS

What is the product of all real solutions of $x^4 - 5x^2 + 4 = 0$?

1. Type `y=x^4-5x^2+4`.

2. Tap the four x-intercepts: $(-2, 0)$, $(-1, 0)$, $(1, 0)$, $(2, 0)$.

3. Multiply them: Type `(-2)(-1)(1)(2)`: it returns 4.

4. Answer: 4.

Answer: 4

34. HARD · DESMOS

A right triangle has legs of length 6 and 8. A larger, similar right triangle has area 150. What is the length of the hypotenuse of the larger triangle?

1. Hypotenuse of the small triangle: Type `sqrt(6^2+8^2)`: it returns 10, and its area Type `6*8/2`: it returns 24.

2. Scale factor: Type `sqrt(150/24)`: it returns 2.5.

3. Hypotenuse of the large triangle: Type `10*sqrt(150/24)`: it returns 25.

4. Answer: 25.

Answer: 25

35. HARD · DESMOS

The graph of the quadratic function f has vertex $(4, -8)$ and passes through the point $(0, 8)$. The function g is defined by $g(x) = f(2x)$. What is the value of $g(1)$?

1. Vertex form $f(x) = a(x-4)^2 - 8$. Use $(0, 8)$: Type `(8+8)/(0-4)^2`: it returns 1 for a.

2. Type `y=(x-4)^2-8` and check it passes through $(0, 8)$: (0, 8).

3. $g(1) = f(2)$: Type `(2-4)^2-8`: it returns -4.

4. Answer: -4.

Answer: -4

36. HARD · DESMOS

The function f is even, so $f(-x) = f(x)$ for every x. The function g is defined by $g(x) = f(x) + f(-x)$. If $f(3) = 7$ and $f(-5) = 2$, what is the value of $g(3) + g(5)$?

1. f is even, so $g(x) = f(x) + f(-x) = 2f(x)$. $g(3)$: Type `2*7`: it returns 14.
2. $f(5) = f(-5) = 2$, so $g(5)$: Type `2*2`: it returns 4.
3. Total: Type `14+4`: it returns 18.
4. Answer: 18.

Answer: 18

37. MEDIUM · NO CALCULATOR NEEDED

A website for a cycling club posted a poll asking, "Should the city spend more on bike lanes?" Of the 1,850 people who chose to respond, 91% voted yes. Which is the main reason the poll result should not be used to estimate the opinion of all city residents?

(A) Respondents selected themselves and are likely to hold similar views. ✓

(B) The sample size is too large.

(C) The question has too few possible answers.

(D) The poll was conducted online, and no online poll can be accurate.

1. How many said yes? Type `0.91*1850`: it returns 1683.5 people (about 1684) voted yes.
2. That is a lot of people, but they chose to respond on a cycling club's site, so they are a voluntary-response sample of people who like bike lanes.
3. A large count does not fix self-selection bias, so the sample-size choice is wrong.
4. Main reason: respondents selected themselves and are likely to hold similar views.

Answer: (A)

38. MEDIUM · NO CALCULATOR NEEDED

A random sample of 500 adults reported their coffee consumption. The adults who drink coffee daily had a lower rate of a certain heart condition than those who do not. Which change to the study would make it possible to conclude that drinking coffee reduces the rate of the condition?

(A) Increase the sample to 5,000 adults.

(B) Include only adults older than 40.

(C) Randomly assign the participants to drink coffee daily or not, then compare the rates. ✓

(D) Ask each adult about more lifestyle habits.

1. Compare the sample sizes: Type `5000/500`: it returns 10 times as many adults would still be an observational study.
2. People chose whether to drink coffee, so other differences (age, lifestyle) could explain the lower rate.
3. Only random assignment balances those other factors between the groups, so only it allows a cause-and-effect conclusion.
4. The change needed: randomly assign the participants to drink coffee daily or not.

Answer: (C)

39. Medium · Desmos

Two independent random samples were taken from the same population to estimate a proportion at the same confidence level. The first sample had 400 members and a margin of error of 6%. The second sample had 1,600 members. Which is most likely the margin of error for the second sample?

(A) 1.5%

(B) 6%

(C) 3% ✓

(D) 12%

1. The margin of error shrinks with $\sqrt{n}$. The sample factor is Type `1600/400` : it returns 4.

2. Its square root: Type `sqrt(1600/400)` : it returns 2.

3. Divide the old margin by it: Type `6/sqrt(1600/400)` : it returns 3 percent.

4. Dividing by 4 instead gives Type `6/4` : it returns 1.5, the trap.

Answer: (C)

40. Medium · Desmos

Of 500 residents surveyed, 240 own a car, 180 own a bicycle, and 70 own both. If one of the surveyed residents is chosen at random, what is the probability that the resident owns neither a car nor a bicycle?

1. Own a car or a bicycle: Type `240+180-70` : it returns 350.

2. Own neither: Type `500-350` : it returns 150.

3. Probability: Type `150/500` : it returns 0.3.

4. Answer: 0.3.

Answer: 0.3

41. Medium · Desmos

In a trial, 90 of 120 patients given Drug X improved, and 60 of 100 patients given Drug Y improved. How many times as likely is a patient given Drug X to improve as a patient given Drug Y?

1. Drug X: Type `90/120` : it returns 0.75.

2. Drug Y: Type `60/100` : it returns 0.6.

3. Ratio: Type `(90/120)/(60/100)` : it returns 1.25.

4. Answer: 1.25.

Answer: 1.25

42. Medium · Desmos

A teacher modeled the final score y of students in a course with the line of best fit $\hat{y} = 82 - 1.6x$, where x is the number of days the student was absent. Which statement is the best interpretation of the number 82 in this model?

(A) A student absent for 82 days is predicted to score 0.

(B) The predicted score decreases by 82 for each day absent.

(C) The mean score of all students in the class is 82.

(D) The predicted score of a student with no absences is 82. ✓

1. Type `y=82-1.6x` and tap where the line crosses the y-axis: (0, 82).
2. Check by typing Type `82-1.6*0` : it returns 82, the predicted score when $x = 0$.
3. x is days absent, so $x = 0$ means no absences. For comparison, Type `82-1.6*5` : it returns 74 is the prediction after 5 absences.
4. The number 82 is the predicted score of a student with no absences.

Answer: (D)

43. Medium · Desmos

A model $y = 0.85x + 12$ was fitted to data in which x ranged from 20 to 80. A student uses the model to predict y when $x = 150$. Which statement about this prediction is best?

(A) It is reliable, because the slope is positive.

(B) It is less reliable, because 150 is outside the range of the data. ✓

(C) It is impossible, because y cannot be computed for $x > 80$.

(D) It is reliable, because the model is linear.

1. Range of the data: Type `0.85*20+12` : it returns 29 to Type `0.85*80+12` : it returns 80 for x from 20 to 80.
2. The prediction at $x = 150$: Type `0.85*150+12` : it returns 139.5.
3. Desmos can compute it, so (C) is false. But 150 lies far outside 20 to 80, so the straight-line pattern may not continue.
4. Answer: it is less reliable because 150 is outside the range of the data.

Answer: (B)

44. Medium · Desmos

The function f is defined by $f(x) = (x - 1)(x - 5)$. The graph of g is the reflection of the graph of f across the y-axis, shifted up 4 units. What is the value of $g(2)$?

1. Reflect across the y-axis and shift up 4: $g(x) = f(-x) + 4$. Type `y=(-x-1)(-x-5)+4` .
2. Read the graph at $x = 2$: (2, 25).
3. Check: Type `(-2-1)(-2-5)+4` : it returns 25.
4. Answer: 25.

Answer: 25

45. Medium · Desmos

Two similar triangles have areas 18 and 50. The altitude to the longest side of the larger triangle is 15. What is the corresponding altitude of the smaller triangle?

(A) 10.8 (B) 5.4

(C) 12.5 **(D) 9 ✓**

1. Length ratio = square root of the area ratio: Type `sqrt(18/50)` : it returns 0.6.

2. Smaller altitude: Type `15*sqrt(18/50)` : it returns 9.

3. Check the area ratio: Type `(9/15)^2` : it returns 0.36 equals Type `18/50` : it returns 0.36.

4. Answer: (D) 9.

Answer: (D)